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compact1, compact2, compact3

java.lang.ref

Class Reference<T>

java.lang.Object
    java.lang.ref.Reference<T>

Direct Known Subclasses:

PhantomReference, SoftReference, WeakReference

```
public abstract class Reference<T>
extends Object
```

Abstract base class for reference objects. This class defines the operations common to all reference objects. Because reference objects are implemented in close cooperation with the garbage collector, this class may not be subclassed directly.

Since:

1.2

Method Summary

Concrete Methods

Modifier and Type	Method	Description
void	clear()	Clears this reference object.
protected Object	clone()	Throws CloneNotSupportedException.
boolean	enqueue()	Clears this reference object and adds it to the queue with which it is registered, if any.

T	get()	Returns this reference object's referent.
boolean	isEnqueued()	Tells whether or not this reference object has been enqueued, either by the program or by the garbage collector.

Methods inherited from class java.lang.Object

equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Method Detail

get

```
public T get()
```

Returns this reference object's referent. If this reference object has been cleared, either by the program or by the garbage collector, then this method returns `null`.

Returns:

`The object to which this reference refers, or null if this reference object has been cleared`

clear

```
public void clear()
```

Clears this reference object. Invoking this method will not cause this object to be enqueued.

This method is invoked only by Java code; when the garbage collector clears references it does so directly, without invoking this method.

isEnqueued

```
public boolean isEnqueued()
```

Tells whether or not this reference object has been enqueued, either by the program or by the garbage collector. If this reference object was not

registered with a queue when it was created, then this method will always return `false`.

Returns:

`true if and only if this reference object has been enqueued`

enqueue

```
public boolean enqueue()
```

Clears this reference object and adds it to the queue with which it is registered, if any.

This method is invoked only by Java code; when the garbage collector enqueues references it does so directly, without invoking this method.

Returns:

`true if this reference object was successfully enqueued; false if it was already enqueued or if it was not registered with a queue when it was created`

clone

```
protected Object clone()
                throws CloneNotSupportedException
```

Throws `CloneNotSupportedException`. A `Reference` cannot be meaningfully cloned. Construct a new `Reference` instead.

Overrides:

`clone` in class `Object`

API Note:

`This method is defined in Java SE 8 Maintenance Release 4.`

Returns:

`never returns normally`

Throws:

`CloneNotSupportedException` - always

Since:

8

See Also:

`Cloneable`

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
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